

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.3: Subspaces

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Definition and first examples

The outputs a matrix can “reach” form what is known as a subspace. The directions a matrix “forgets” also form a subspace. The definitions below make this precise.

We are already familiar with geometric spaces living inside bigger ones, like lines lying in the plane, or planes lying in 3-space. We are now going to formalize this notion algebraically. Recall the definition of the set $\mathbb{R}^n$ in Vector.

Definition: Subspaces

A set U of vectors in $\mathbb{R}^n$ is called a of $\mathbb{R}^n$ if it satisfies the following properties:

- The zero vector (written as $\mathbf{0}$) is in U .
- If $\mathbf{x}$ is in U and $\mathbf{y}$ is in U , then $\mathbf{x} + \mathbf{y}$ is in U . This means U is *closed under addition*.
- If $\mathbf{x}$ is in U , then $a\mathbf{x}$ is in U for every scalar a . This means U is *closed under scalar multiplication*.

Note: Subspaces are closed under linear combinations

Let U be a subspace. If $\mathbf{u}, \mathbf{v} \in U$ and $a, b \in \mathbb{R}$, closure under scalar multiplication gives $a\mathbf{u} \in U$ and $b\mathbf{v} \in U$. Closure under addition then gives

$$a\mathbf{u} + b\mathbf{v} \in U.$$

More generally, if $\mathbf{u}_1, \dots, \mathbf{u}_k \in U$ and $c_1, \dots, c_k \in \mathbb{R}$, each vector $c_i \mathbf{u}_i$ lies in U . Adding these vectors one at a time shows that

$$c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k \in U.$$

Thus every finite linear combination of vectors in a subspace remains in that subspace.

Example

The whole set $\mathbb{R}^n$ is a subspace of itself.

Example

The set containing only the zero vector, i.e., the set $\{\mathbf{0}\}$, is a subspace of $\mathbb{R}^n$ (called the or).

Activity: The zero vector and subspace conditions (U2-L04)

Verify that $\{\mathbf{0}\}$ is a subspace of $\mathbb{R}^n$.

Definition: Proper Subspaces

Any subspace of $\mathbb{R}^n$ other than $\mathbb{R}^n$ or $\{0\}$ is a
of $\mathbb{R}^n$.

Example

Planes and lines through the origin in $\mathbb{R}^3$ are all subspaces of $\mathbb{R}^3$.

Example: Lines and planes through the origin (U2-LO4)

Task

Verify that planes through the origin in $\mathbb{R}^3$ are all proper subspaces of $\mathbb{R}^3$.

Task

Verify that lines through the origin in $\mathbb{R}^3$ are subspaces of $\mathbb{R}^3$.

Note

Lines and planes which do not pass through the origin in $\mathbb{R}^3$ are *not* subspaces of $\mathbb{R}^3$. Nonetheless, if Π is a plane, and we consider the set of displacement vectors

$$\Delta\Pi = \{\mathbf{v}_1 - \mathbf{v}_0 : \mathbf{v}_0, \mathbf{v}_1 \in \Pi\},$$

then $\Delta\Pi$ forms a plane passing through the origin, and so is a subspace of $\mathbb{R}^3$. A similar result holds for sets of displacement vectors given by a line not passing through the origin.

Example: A product condition on coordinates (U2-LO4)

Show that the set

$$U = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 x_2 = 0 \right\}$$

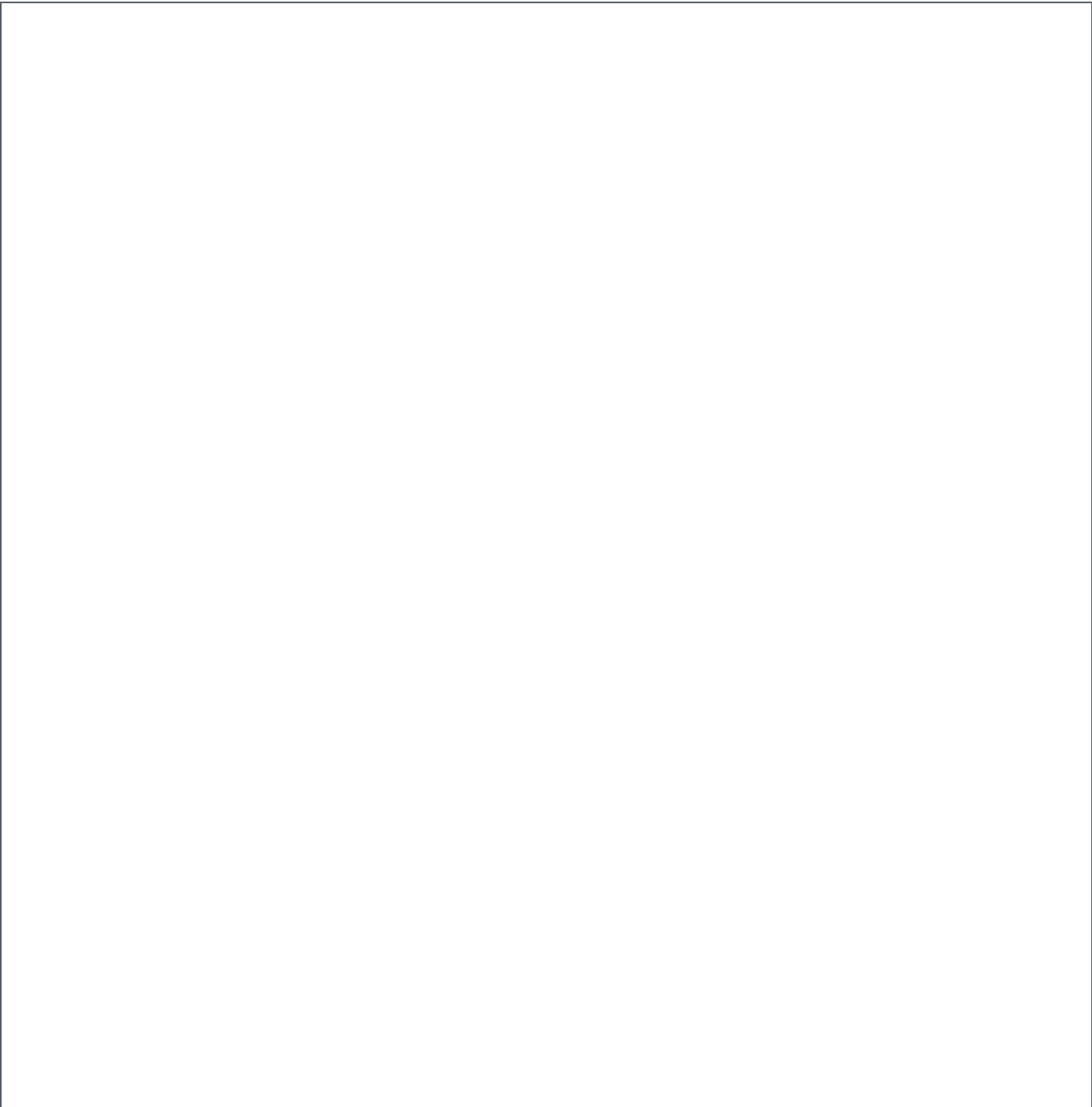
is *not* a subspace of $\mathbb{R}^2$.

Activity: A set described by two parameters (U2-LO4)

Let W be the set of all vectors in $\mathbb{R}^3$ of the form

$$\begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix},$$

where $r, s \in \mathbb{R}$. Is W a proper subspace of $\mathbb{R}^3$?



Note

To show a set *is* a subspace, you tend to need to parameterize all elements of the set in order to show they all satisfy some property (closure under addition or scalar multiplication), and so your argument should use variables. To show a set *is not* a subspace, you need to find one or more counterexamples that show that some property does not hold for all elements; a single explicit counterexample usually suffices.

Null space and image space

The following important examples are representative of two canonical ways of describing a subspace.

Definition: Null Space and Image Space

Let A be an $m \times n$ matrix. The of A , denoted $\text{null}(A)$, is the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ which are solutions to the equation $A\mathbf{x} = \mathbf{0}$, i.e.,

$$\text{null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}.$$

The of A , denoted $\text{im}(A)$, is the set of vectors $\mathbf{y}$ in $\mathbb{R}^m$ such that $A\mathbf{x} = \mathbf{y}$ has a solution, i.e.,

$$\text{im}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}.$$

Note

The null space lives in the input space. It records input directions that A sends to $\mathbf{0}$.

The image space lives in the output space. It records outputs that A can reach.

- **Object:** $\text{null}(A)$
Lives in: $\mathbb{R}^n$
How to read it: input directions A sends to $\mathbf{0}$
- **Object:** $\text{im}(A)$
Lives in: $\mathbb{R}^m$
How to read it: outputs A can reach

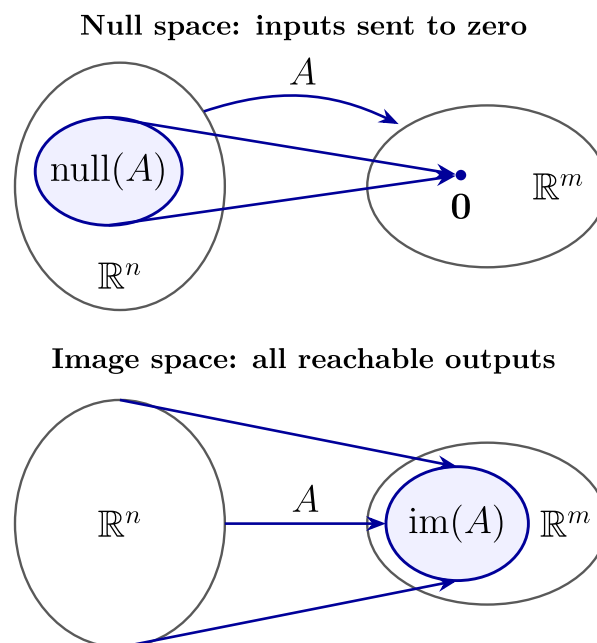


Figure. The null space consists of inputs sent to the zero vector. The image consists of all outputs reached from the input space.

Activity: Revisiting the height map with subspace language (U2-LO3)

Return to A map that forgets height (U2-LO1, U2-LO3). Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Describe the output subspace and the subspace of input directions that P sends to $\mathbf{0}$.

Theorem

For any $m \times n$ matrix, the null space of A is a subspace of $\mathbb{R}^n$, and the image space of A is a subspace of $\mathbb{R}^m$.

Why is this true?

We start by showing the null space of A is a subspace. Recall the properties discussed in Properties of matrix-vector products:

Contains the zero vector $\mathbf{0} \in \text{null}(A)$ because $A\mathbf{0} = \mathbf{0}$.

Closure under addition

Suppose $\mathbf{v} \in \text{null}(A)$ and $\mathbf{w} \in \text{null}(A)$. Then $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{w} = \mathbf{0}$. So

$$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Thus $\mathbf{v} + \mathbf{w} \in \text{null}(A)$.

Closure under scalar multiplication

If $\mathbf{v} \in \text{null}(A)$, and a is a scalar, then

$$A(a\mathbf{v}) = a(A\mathbf{v}) = a\mathbf{0} = \mathbf{0}.$$

Thus $a\mathbf{v} \in \text{null}(A)$.

So $\text{null}(A)$ is a subspace of $\mathbb{R}^n$.

Next, we show that $\text{im}(A)$ is a subspace.

Contains the zero vector $\mathbf{0} \in \text{im}(A)$ because $A\mathbf{0} = \mathbf{0}$.

Closure under addition

Suppose $\mathbf{v} \in \text{im}(A)$ and $\mathbf{w} \in \text{im}(A)$. Then the equations $A\mathbf{x} = \mathbf{v}$ and $A\mathbf{x} = \mathbf{w}$ both have solutions, i.e., there exists $\mathbf{x}_1$ and $\mathbf{x}_2$ such that $A\mathbf{x}_1 = \mathbf{v}$ and $A\mathbf{x}_2 = \mathbf{w}$. Then

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{v} + \mathbf{w},$$

so the equation $A\mathbf{x} = \mathbf{v} + \mathbf{w}$ has a solution, and so $\mathbf{v} + \mathbf{w} \in \text{im}(A)$.

Closure under scalar multiplication

If $\mathbf{v} \in \text{im}(A)$, and a is a scalar, then we may write $\mathbf{v} = A\mathbf{x}_1$ for some $\mathbf{x}_1$. But then

$$A(a\mathbf{x}_1) = a(A\mathbf{x}_1) = a\mathbf{v},$$

so the equation $A\mathbf{x} = a\mathbf{v}$ has a solution, so that $a\mathbf{v} \in \text{im}(A)$.

Thus $\text{im}(A)$ is a subspace of $\mathbb{R}^m$.

Warning

The solution set of $A\mathbf{x} = \mathbf{0}$ is a subspace. The solution set of $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} \neq \mathbf{0}$ is usually not a subspace; when it is nonempty, it is a shifted copy of $\text{null}(A)$.

Fact: General solution of a linear system

Let A be an $m \times n$ matrix. If $\mathbf{x}_p$ is one solution of $A\mathbf{x} = \mathbf{b}$, then

$$\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{b}\} = \mathbf{x}_p + \text{null}(A).$$

That is, the solutions are exactly the vectors $\mathbf{x}_p + \mathbf{z}$ with $\mathbf{z} \in \text{null}(A)$.

Why is this true?

Since $A\mathbf{x}_p = \mathbf{b}$, for any $\mathbf{x} \in \mathbb{R}^n$,

$$\begin{aligned} A\mathbf{x} = \mathbf{b} &\iff A(\mathbf{x} - \mathbf{x}_p) = \mathbf{0} \\ &\iff \mathbf{x} - \mathbf{x}_p \in \text{null}(A). \end{aligned}$$

Thus $A\mathbf{x} = \mathbf{b}$ exactly when $\mathbf{x} = \mathbf{x}_p + \mathbf{z}$ for some $\mathbf{z} \in \text{null}(A)$.

From here on, homogeneous equations play a special role. To solve $Ax = \mathbf{0}$ by row reduction, we could write the augmented matrix $[A \mid \mathbf{0}]$. Since the right-hand side stays zero throughout the row operations, the coefficient matrix A carries the essential information.

Spanning Sets

Even though subspaces of $\mathbb{R}^n$ can have infinitely many vectors, it is possible to represent them with a finite amount of data.

Activity: Writing a parametric set as a span (U2-LO4)

Recall from A set described by two parameters (U2-LO4) that

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}.$$

Show that there are two vectors (describe them explicitly), so that every vector in W can be written as a linear combination of those two vectors.

Definition: The Span of a Set of Vectors

The set of all linear combinations of a set of vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ in $\mathbb{R}^n$ is

called the of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$, and is denoted

$$\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \{t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k : t_1, \dots, t_k \in \mathbb{R}\}.$$

If $V = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, we say that V is by the vectors

$\mathbf{x}_1, \dots, \mathbf{x}_k$, and that the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ the space V .

Example

Recall the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$ from Standard basis. Note that if

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

then

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \dots + x_n\mathbf{e}_n.$$

This means that $\mathbb{R}^n = \text{span}\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.

For an image space, spanning vectors come from the columns of the matrix. If

$$A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$$

and

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Thus every output $A\mathbf{x}$ is a linear combination of the columns of A .

Activity: A spanning set for an image space (U2-LO3, U2-LO4)

Find a spanning set for the image space of the matrix

$$A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}.$$

Activity: A spanning set for a null space (U2-LO3, U2-LO4)

Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}.$$

Practice this activity in [The Elimination Scratchpad](#)



To summarize: basic solutions of $A\mathbf{x} = \mathbf{0}$ form a spanning set for $\text{null}(A)$, and the columns of A form a spanning set for $\text{im}(A)$.

Theorem

Let $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ be a set in $\mathbb{R}^n$. Then

- U is a *subspace* of $\mathbb{R}^n$ containing each of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$.
- If W is a subspace of $\mathbb{R}^n$ and each of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ is in W , then U is a *subset* of W , i.e., $U \subseteq W$.

Why is this true?

Define an $n \times k$ matrix A , with the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as columns. Then $U = \text{im}(A)$, and is thus a subspace of $\mathbb{R}^n$. Since $\mathbf{x}_i = A\mathbf{e}_i$ for $1 \leq i \leq k$, the set U contains each of the vectors $\mathbf{x}_i$.

Conversely, suppose W is a subspace of $\mathbb{R}^n$ and $\mathbf{x}_1, \dots, \mathbf{x}_k \in W$. Since

W is closed under addition and scalar multiplication, all of the linear combinations of $\mathbf{x}_1, \dots, \mathbf{x}_k$ are elements of W , and so $U \subseteq W$.

Activity: Testing membership in a span (U2-LO4)

In $\mathbb{R}^3$, consider the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}.$$

Determine if the vector

$$\mathbf{w} = \begin{bmatrix} 1 \\ 5 \\ -7 \end{bmatrix}$$

belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.

Activity: Testing membership in another span (U2-LO4)

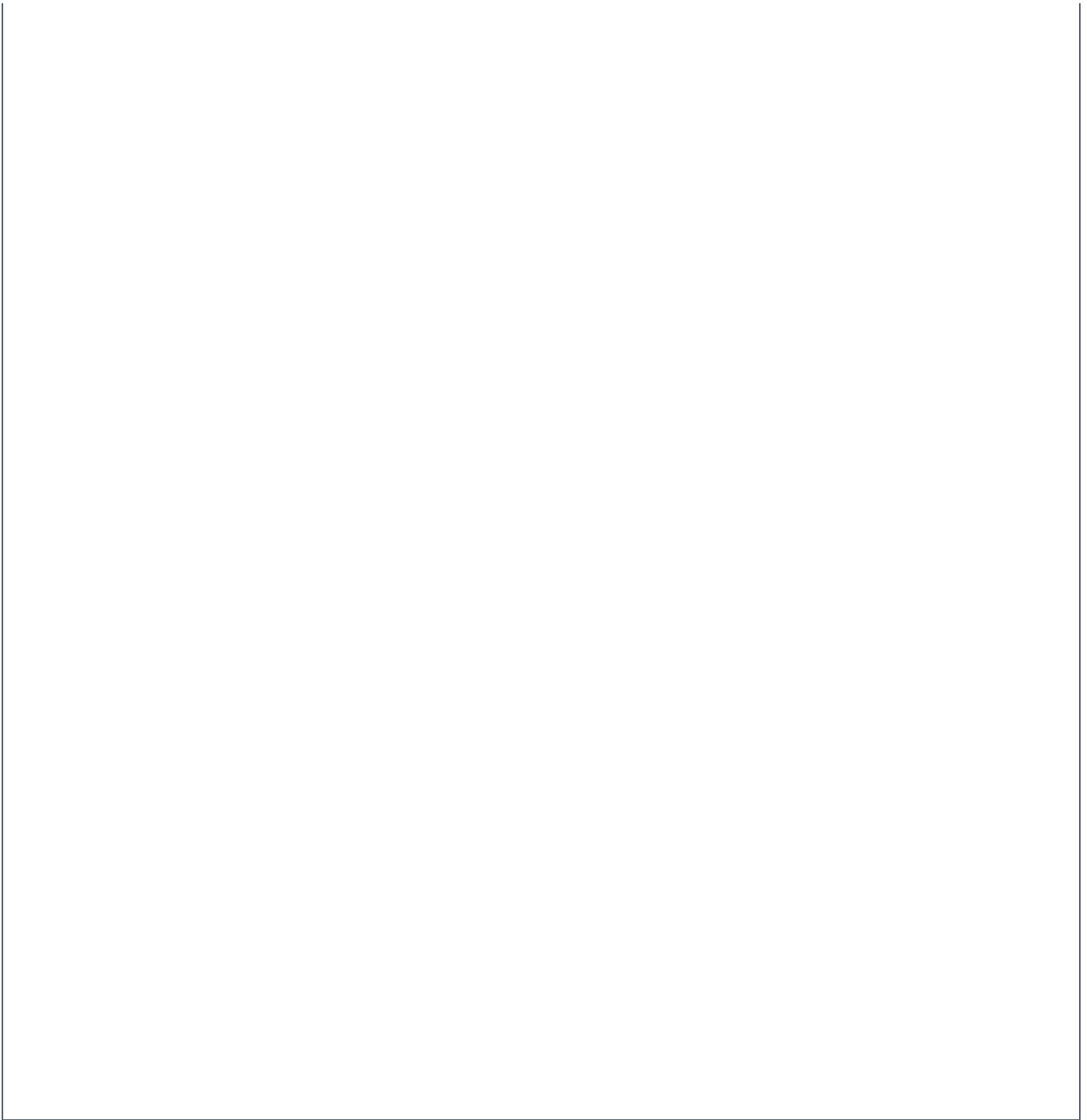
In $\mathbb{R}^3$, consider the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

Determine if the vector

$$\mathbf{w} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.



Fact: Spanning and solvability

Let A be an $m \times n$ matrix with columns $\mathbf{c}_1, \dots, \mathbf{c}_n$. Then $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^m$ if and only if $\text{span}\{\mathbf{c}_1, \dots, \mathbf{c}_n\} = \mathbb{R}^m$.

Why is this true?

As noted in the earlier discussion of matrix columns,

$$A\mathbf{x} = x_1\mathbf{c}_1 + \dots + x_n\mathbf{c}_n.$$

Thus $A\mathbf{x} = \mathbf{b}$ has a solution exactly when $\mathbf{b}$ is a linear combination of the columns of A . By The Span of a Set of Vectors, this holds for every $\mathbf{b} \in \mathbb{R}^m$ exactly when the columns span $\mathbb{R}^m$.

The next section studies when a spanning set has redundant vectors and how to remove them.